

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0873

Omni-Hackberry 230 kV
January 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0873. The Interconnection Customer proposes a 473.3 MW natural gas combined cycle facility interconnecting at the Omni-Hackberry 230 kV substation in the Ohio area. The request was received on October 08, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 359 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,321,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Omni-Hackberry Substation 500 kV/230 kV XFMR: 113.6% loading (1,409 MVA) following the outage of the Omni-Hackberry-Norwich Substation 500.0 kV L; DFAX = 43.1%.

The Capitol Heat And Power-International Paper 230.0 kV is overloaded to 105.5% of its 535 MVA emergency rating for the loss of the Omni-Hackberry 500 kV/230 kV TX. The project contributes 23.5% DFAX (111.0 MW) to this overload.

For the loss of Beck Rd-Dillard-Orange 345.0 kV, the Cairo-Sporn 345.0 kV loads to 102.8% of its 1,220 MVA rating. Project DFAX: 5.5%.

For the loss of Omni-Hackberry Substation 500/230 kV TX, the Capitol Heat And Power-Omni-Hackberry 230.0 kV loads to 100.2% of its 609 MVA rating. Project DFAX: 34.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Omni-Hackberry Substation 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$26,304,000	17% (\$4,360,000)
Capitol Heat And Power-International Paper 230.0 kV	Reconductor line	\$7,185,000	26% (\$1,830,000)
Cairo-Sporn 345.0 kV	Reconductor line	\$45,145,000	5% (\$2,455,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Capitol Heat And Power-Omni-Hackberry 230.0 kV	Reconductor line	\$11,460,000	23% (\$2,676,000)
	Total allocated to project		\$11,321,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.